

OSPO@GT Update

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Principal Research Scientist and Senior Research Scientist, OIT/PACE
Co-Directors @ GT OSPO

OSPO@GT - Spring 26 Maintainers Meeting
January 13th, 2026

Mentimeter Survey

Go to www.menti.com

Enter the code **6585 9701**

Share some of your interests and thoughts about how you are engaging with open source!



Georgia Tech's OSPO aims to build open-source capacity on campus and at nearby academic institutions through education on best practices, outreach, and training and engagement of students.

OSPO@GT Core Objectives – 2025-2027

OSPO Goal 1: Support GT OSPO Sustainability

Improve OSS Developer Engagement

Create a welcoming community for all developers and engage in robust community outreach

Develop added metrics to track OSPO progress and criteria for success

Support Developers at GT and other Atlanta universities through workshops, training sessions, and meet-ups

Institutionalize Student Programs

Evolve the virtual summer internship program to be more sustainable and allow for more opportunity for all interested students

Enhance and Improve training materials to allow for more large-scale, asynchronous OSS training programs

Address the AI Revolution

Educate students on the benefits and pitfalls of open-source AI and current viewpoints, including the ethical use of AI

Host Open Data Sets and exploration tools for AI data agents

Provide training for AI hardware and software tools on local HPC resources

Design joint software engineering training with other GT groups

Explore GT Strengths in OSS

Collaborate with Georgia Tech's School of Cybersecurity and GTRI to develop and share secure open-source best practices.

Support institute efforts in maintaining research integrity using open-source techniques

OSPO Goal 2: Support a Sustainable OSS Community for GT and Atlanta

Core Team Members



Jeff Young

PI, Director



Fang (Cherry) Liu

Co-PI, Associate Director



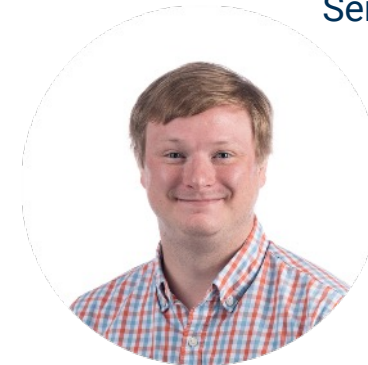
Ron Rahaman

Senior Personnel



Cliff Landis

Digital Curation Archivist



Dillon Henry

Digital Accessioning Archivist

***Joint effort among
COC, OIT/PACE and
Library***

Advisory Board Members



David Sherrill

Regents' Professor, IDEAS
Interim Director



Suresh Marru

Principal Research Scientist
Director, Center for AI in
Science and Engineering
(ARTISAN)
Institute of Data Engineering
and Science (IDEaS)



Eric Sembrat

Director of Digital Learning
Technologies



Nicholas Bollweg

Senior Research Engineer
GTRI-ELSYS

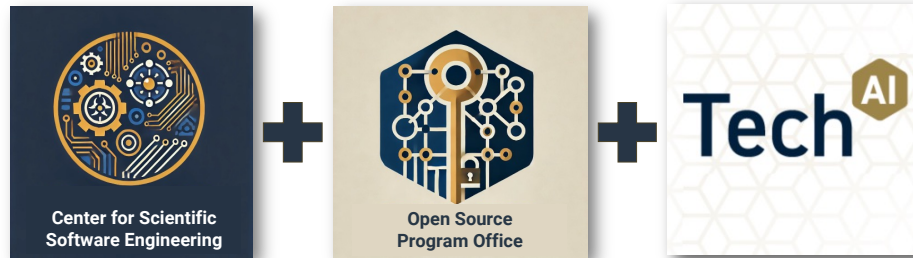
We are actively looking for additional advisory board members!

Connections between OSPO@GT and Research Software

The creation of Georgia Tech's Center for Scientific Software Engineering (CSSE) led to the application for and award of a Sloan-funded Open Source Program Office at Georgia Tech

Successful open source projects and ecosystems require good software engineering practices!

- We believe that a monthly maintainers meeting is also a good avenue to connect with GT's **research software engineers (RSEs)**.



2025 OSS and Scientific Software Workshop
<https://ospo.cc.gatech.edu/ospo-sse-workshop-2025/>

- We anticipate using these meetings to connect concepts useful both to RSEs and open source practitioners!

What will we do in these meetings?

Training and Best Practices

- Talk about recent tools and training that the OSPO has been developing

Lessons Learned

- We strongly encourage attendees to bring lightning talks on their work to future meetings. Encourage participation in your open source project!

Discussion

- Is there a hot topic that we should all cover? We can discuss at a high level and then break out into small groups to go deeper

OSPO@GT Activities

- Licensing Standards and Workflows
 - Released licensing workflow and guidelines on open sourcing projects at GT
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 - General meetings with AE, Chemistry, School of Music, GTRI, CoC, and many others!
- Student engagement and training
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GT OSS Policies - Interaction with Office of Technology Licensing

- GT's Office of Technology Licensing (OTL) assists researchers with licensing and commercializing R&D. All software can receive copyright, and **some** software may be patentable.
- GT employees and those working on sponsored research are required to disclose their software creations to OTL for evaluation via a software disclosure form¹
- One outcome of OTL's internal software disclosure process can be release via an open source license.
- The OSPO supports the OTL when a researcher decides to open source their code.

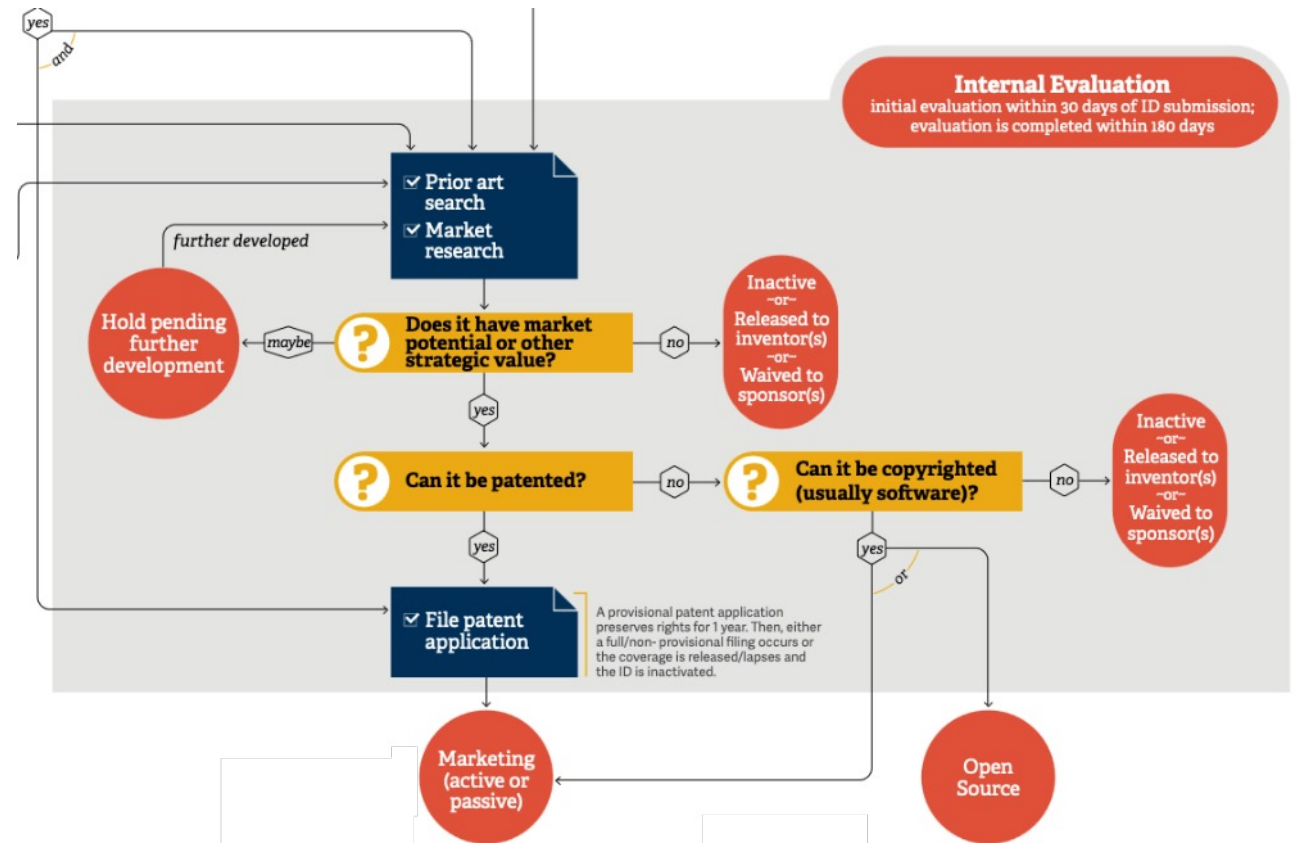
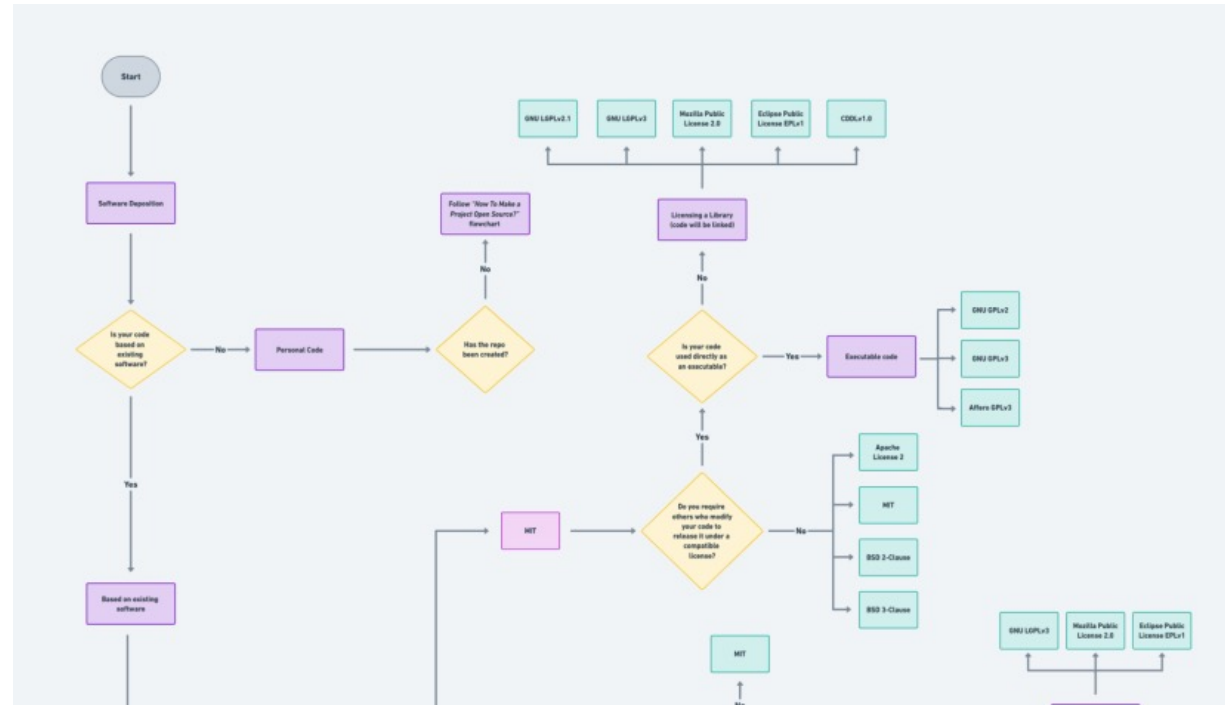


Figure Excerpt from: <https://licensing.research.gatech.edu/sites/default/files/gt-otl-inventiondisclosure-flowchart.pdf>

GT OSPO Licensing Workflows



Resources on the GT OSPO website include flowcharts to allow for future exploration of different common license options for software and data artifacts.

- Software license selection
<https://ospo.cc.gatech.edu/open-source-software-licensing/>
- Data license selection
<https://ospo.cc.gatech.edu/open-source-data-licensing/>

GT OSPO Licensing Guidelines

Some OSS developers just want a simple guide that suggests compatible OSS licenses and pros/cons for permissive versus copyleft licenses!

Questions to Ask When Licensing Your Code

1. What are the requirements of your funding for open sourcing and open source licenses?
 - Some sponsors like NSF and [NIH](#) encourage open source releases of software development.
2. Who is the target audience for your code? Do they normally use a specific license?
 - For example, the widely used [NumPy project](#) uses a [modified BSD](#) license, so related work may also use a similar license.
3. How do you want others to contribute back to your codebase?
 - Do you want to include as many contributors as possible? Do you want to restrict contributions to certain types of products?
 - Answering these questions may determine whether you select a permissive or less permissive license.
4. Do you prefer contributions from government and industry partners?
 - US government and many industry partners may only be able to contribute to a project under certain licenses.

License Recommendations for Software

Preferred: [MIT License](#), [BSD License](#), [Apache 2.0 License](#), [LGPL v3.0 License](#)

See [TL;DR Legal](#) for a summary of these licenses. You can also use [Choose A License](#) to evaluate your options.

Read more at <https://ospo.cc.gatech.edu/releasing-open-source/>

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OSPO@GT - Virtual Summer Internship (Summer 2024-2026)

Virtual summer internship program (VSIP) runs for 10 weeks from mid-May to the end of July

- 12 open-source projects were chosen with GT and IBM mentors
- 2-3 students per project
- Weekly training sessions, meetings with mentors, and a final poster session
- VSIP 2025 program hosted ~30 students

Projects included CFD codes, nuclear fusion simulations, container solutions, and open source AI training

Learn more at <https://ospo.cc.gatech.edu/vsip/>

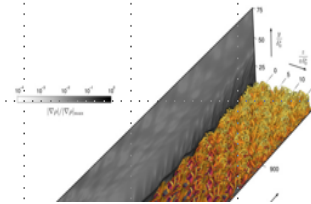
- **Contribution** to Prompt Accessibility, Maintainability, and Rapid Development.

Low Stakes Issues

- CPU/GPU Profiling: *Perf* & *Nsys* to highlight bottlenecks.
- Containerization: *Apptainer* for MFC with all essential packages/compilers on OSG Access Point.
- Continuous Benchmarking: Performance improvement of releases.
- GitHub CI: Source Code Analyzer (*PMD*); Frontier benchmark case; RMDA MPI cases, etc.

- **Goal** Transitional Boundary Layer Case

- Study: Non-Equilibrium Gas Dissociation for 7-Species Model on Flat Plate.
- Assessment: Riemann Solvers to Capture Aerothermodynamic Phenomena.
- Discovery: Exascale Parallelization Framework (Legion) Vs. Standard Use of OpenMP/OpenACC.
- Exploration: Potential Use of MFC to Design Thermal Protection Systems (TPS).



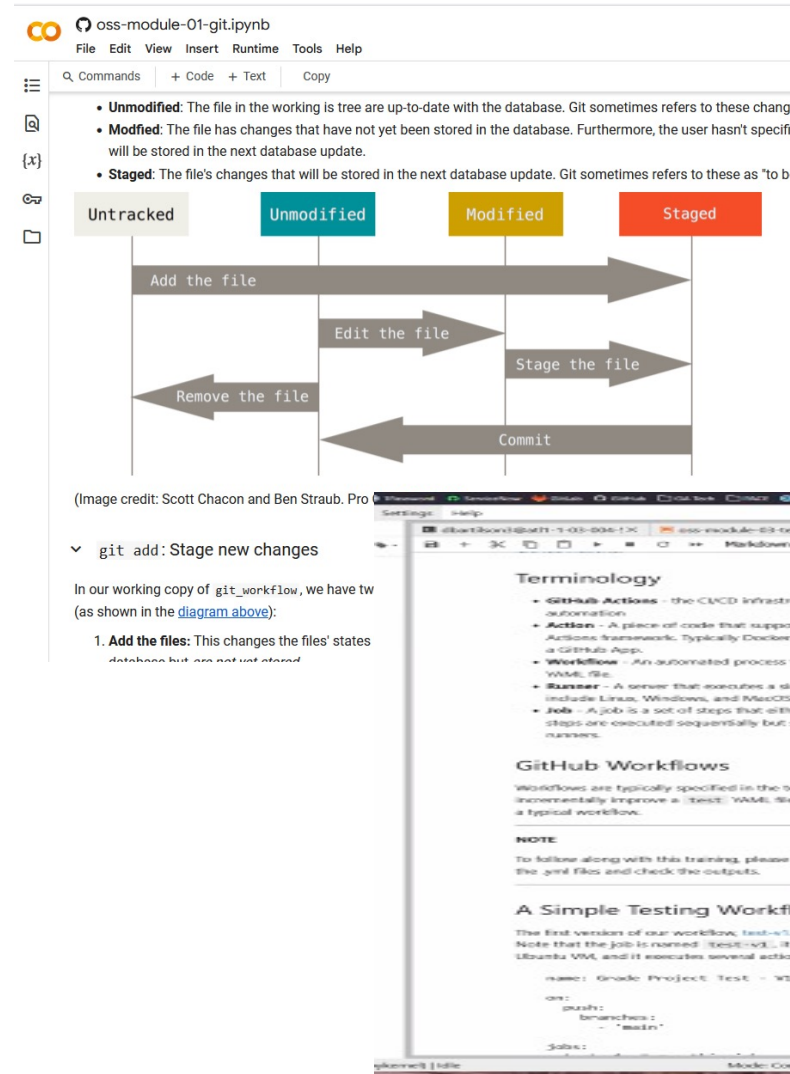
Student update from MFC project,
<https://mflowcode.github.io/>

MFC was a Gordon Bell Finalist in 2025!

OSPO@GT - OSS Training

8 lessons on open source training topics developed with PACE and CSSE

- Git – forks, issues, PRs
- Testing Code
- CI/CD
- Reviewing Code
- OSS Governance Models
- OSS Project Lifecycles



Training Lecture on Testing by Daniel Bartilinson

Learn more at <https://github.com/gt-ospo/oss-training>

Enhancing our Asynchronous Training Capabilities

Key challenge:

GT has a large study body; how can we teach all of our students key open source concepts?

Asynchronous training certificate program in OSS basics

- Focusing on mapping existing OSS training to open source nbgrader framework
- Currently working on Git basics but anticipate added OSS classes, software engineering

```
Before you turn this problem in, make sure everything runs as expected. First, restart the kernel (in the menubar, select Kernel → Restart) and then run all cells (in the menubar, select ).

Make sure you fill in any place that says YOUR CODE HERE or "YOUR ANSWER HERE", as well as your name and collaborators below:

[ ]: NAME = ""
COLLABORATORS = ""

Checkout commit that goes back to Lesson 2 start point

[1]: import os
from git_state_manager import save_git_state, load_git_state
os.chdir("oss-training-templates")

[2]: def checkout_commit():
    # YOUR CODE HERE
    command = !git checkout d4034cff631ebda366cb68353368dde7eb25c32

    return command

print(checkout_commit())
print("-----\n" + save_git_state("git_state_2.2.json", "test_1"))

["Note: switching to 'd4034cff631ebda366cb68353368dde7eb25c32'.", '', "You are in 'detached HEAD' state. You can look around, make experimental", 'changes and com', 'n discard any commits you make in this', 'state without impacting any branches by switching back to a branch.', '', 'If you want to create a new branch to retain c', 'ou may', 'do so (now or later) by using -c with the switch command. Example:', '', ' git switch -c <new-branch-name>', '', 'Or undo this operation with:', '', 'Turn off this advice by setting config variable advice.detachedHead to false', '', 'HEAD is now at d4034cf First commit of my_abs (no try/except yet)']
-----
Saved git state for test_1.

[3]: git_state = load_git_state("git_state_2.2.json", "test_1")

expected_commit = "d4034cff631ebda366cb68353368dde7eb25c32"
assert git_state["commit"] == expected_commit, f"Expected commit '{expected_commit}', found '{git_state['commit']}'"

[4]: def checkout_main():
    # YOUR CODE HERE
    command = !git checkout main

    return command
```

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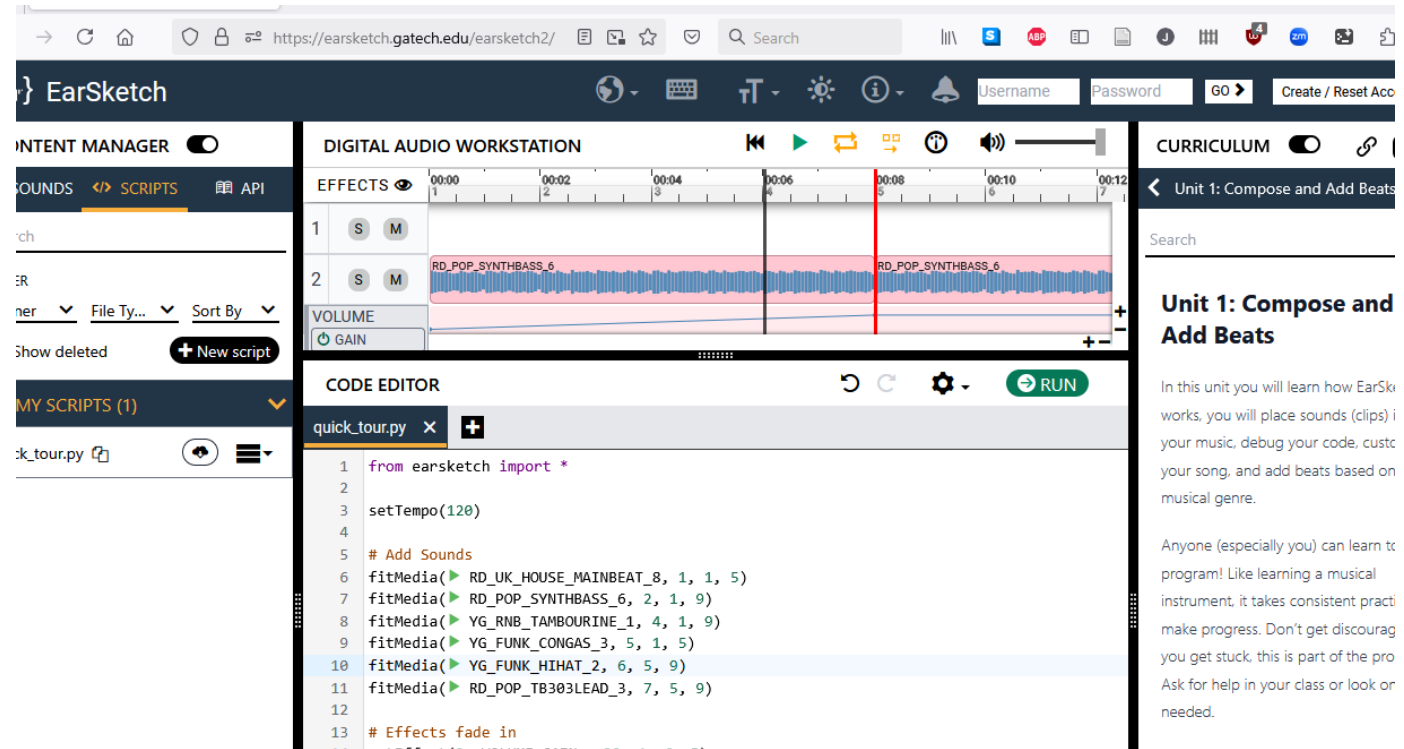
Funding Opportunities for OSS

Some limited opportunities for funding OSS from the OSPO

- VSIP program, faculty fellowships
- 5 fellowships awarded in Spring 2025

More importantly, the OSPO can serve to support and connect groups with funding opportunities!

- 6 GT proposals to 2024 NSF POSE call
 - **At least one confirmed awardee for 2025 - Psi4!**
- NSF CSSI grants
- CZI
- NASA, MoISSI calls



Earsketch, a 2024 POSE Phase 1 Awardee.

Learn more at <https://ospo.cc.gatech.edu/oss-funding-opportunities/>

OSPO@GT Activities

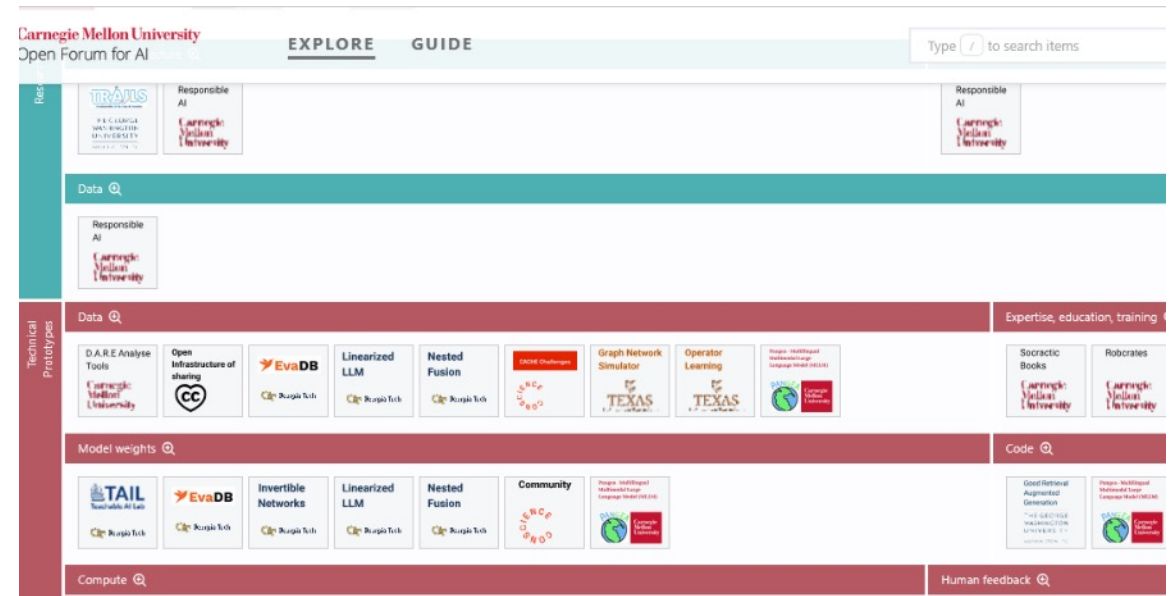
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Open Forum for AI (OFAI) and OSI Definition

Open Forum for AI

- GT joined this CMU-led effort in 2024
- Provides a balanced voice focused on transparent, responsible, safe and ethical AI
- A recent effort on identifying the AI landscape includes five GT research groups

<https://openforumai.github.io/landscape/>



DONATE About Open Source Definition Licenses Open Source AI

What is Open Source AI

When we refer to a “system,” we are speaking both broadly about a fully functional structure and its discrete structural elements. To be considered Open Source, the requirements are the same, whether applied to a **system**, a **model**, **weights** and **parameters**, or other structural elements.

An *Open Source AI* is an AI system made available under terms and in a way that grant the freedoms¹ to:

- **Use** the system for any purpose and without having to ask for permission.
- **Study** how the system works and inspect its components.
- **Modify** the system for any purpose, including to change its output.
- **Share** the system for others to use with or without modifications, for any purpose.

OSI AI 1.0 definition

- Aims to define and promote open-source standards related to transparency, innovation, and equitable access around open-source software and data.

<https://opensource.org/ai/open-source-ai-definition>

CURIOSS

The CURIOSS meta-organization hosts monthly meetings, in-depth seminars, and biyearly in-person meetings

- Provides a useful gathering of open source resources and groups at <https://curioss.org/resources/>
 - This can be a great place to find open source tools and best practices!
- GT is one of the founding members for this group

Recent Focus of CURIOSS meetings – OSPO Patterns

- Patterns that provide ***specific examples*** of activities that OSPOs are working on
- Useful for prioritizing activities an OSPO participates in



Open Source Project Explorer

An open source GitHub Pages web interface that allows GT researchers:

- To share their open source projects and relevant details
- To explore new projects, licenses, and tools
 - Allows faculty to make new connections with their research
 - Enables students to learn about and contribute to open source projects
- Helps the OSPO to do **community discovery**

Open Source Projects

Search project by name...

About This Resource

This page provides an overview of Georgia Tech open source research projects curated by the Georgia Tech Open Source Project Office (OSPO). Georgia Tech PIs and researchers can submit their work via the form below, and all submissions are reviewed via a PR process before being posted.

If you have any questions about this resource, please reach out to us via email at ospo-team@gatech.edu.

Project Area

☐ Artificial Intelligence

☐ Bioscience

☐ Computer Science

☐ High Performance Computing

☐ Computer Graphics

☐ Robotics

☐ Human-Computer Interaction

License

☐ MIT License

☐ Apache License 2.0

☐ ISC License

☐ BSD 3-Clause "New" or "Revised" License

Name	Project Areas	Licenses
> Tech-Matrix	CS, Robotics, AI, HCI	ISC
▼ Quantum-Hub	Graphics, Bioscience, HPC, AI	Apache, MIT
Abstract Innovating at the confluence of Graphics, HPC, AI, Bioscience while emphasizing to address critical challenges.	Primary Contact(s) 1. Quinn Miller (qmiller@example.net)	URLs Project URL: http://emdhsobtx-tech.com/ Guidelines URL: http://fnvkoh-tech.com/
> Neural-Forge	Bioscience, HCI	BSD-3
> Cyber-Flow	CS	ISC, BSD-3
▼ Tech-Matrix	CS, HPC, AI	Apache, MIT
Abstract Innovating at the confluence of CS while emphasizing to transform and revolutionize the field.	Primary Contact(s) 1. Jamie Taylor (jtaylor@example.com)	URLs Project URL: http://kkaapface-tech.com/ Guidelines URL: http://pdhnb-innova.com/
> Quant-Sys	Graphics, HPC, CS, Bioscience	BSD-3
> Eco-Forge	HCI, AI, Bioscience, Robotics	MIT, ISC
> Eco-Hub	HCI, Graphics, Bioscience, HPC	ISC, MIT
> Nano-Flow	CS, Robotics	MIT
> Cloud-Flow	HPC, HCI, CS	BSD-3

Open Source Project Explorer – Related Efforts

 **LibreLinker**
Connecting Ideas, Building Tomorrow, Fighting for Freedom

Discover Free Software & Hardware Projects

A curated collection of innovative open projects spanning AI, chemistry, hardware, and beyond – all seeking to respect your freedom. All projects **respect your freedom** and are GPL compatible. Most all are led by Georgia Tech students, faculty, and staff. The **LibreTech Collective**, Georgia Tech's only Free, Libre, & Open-Source club, invites you to explore, contribute, and make a difference.

Thanks to **GT-OSPO** for helping assemble parts of this list.

Filter by type: AI Academic Research Hardware Web Enterprise Plugin

Project	Description	Type	Technologies	Year
 CAS	CAS (Central Authentication Service) is a multilingual single sign-on solution and open-source identity provider for enterprise authentication and authorization. It provides...	 	SSO Authentication +3	2004
 TrustGraph	TrustGraph eliminates AI hallucinations by ingesting structured and unstructured data into private knowledge graphs with deep data relationships. It provides GraphRAG...	  	GraphRAG Knowledge Graphs +3	2024

LTC@GT has a similar tool at <https://librelinker.us/> that focuses on projects students can engage with.

LTC@GT – Student Run Open Source Club!

LibreTech Collective is a new club focused on open source hacking for software, hardware, and knowledge!

- 🛠 Hands-on Projects: Build free software and hardware from the ground up, from anywhere in the world.
- 🌐 Upstream Contributions: Contribute to global open-source projects like Linux, GNU, Apache, and more.
- 🎉 Events & Sprints: Join hackathons, code sprints, workshops, and live-speaker events.
- 📢 Advocate for Freedom: Raise awareness about software freedom, copyright, and inclusive tech practices.
- 📖 Learn & Teach: Learn by contributing, mentor others, and grow as a technologist.



GEORGIA TECH
LIBRETECH COLLECTIVE

It's GNU for you - Hack. Share. Learn.

Learn more at <https://ltc.gtorg.gatech.edu/> or <https://github.com/LTC-GT>

OSPPO Opportunities for Engagement

Virtual Summer Internship Program for students in Summer 2026

- Similar to Google Summer of Code with focus on training students to contribute to open source projects
- Long-term goal of integrating with VIP or UROC programs

Engage the OSPPO with your open source questions

- What requirements does NSF funding have for open sourcing software?
- What if I want to both patent and open source my research software?
- Does Georgia Tech have an official open source license we should use?
- Should I open source data used for my new AI model?

Look out for Funding Notices for OSS Research and Development

- Faculty fellowships, POSE, CZI, Apereo, etc.

Training

- Open source AI topics
- Software Preservation
- Securing Open Source

Connect with OSPO@GT

- Email Lists, MS Teams group
- Discord (student-focused)
- LinkedIn and BlueSky



Email List

<https://bit.ly/3YBvZKJ>



MS Teams

<https://bit.ly/3LdvJP4>



Discord

<https://discord.gg/x73GXzPPH6>